



uOttawa

Hong et al. (2022). Mitochondrial dynamics maintain muscle stem cell regenerative competence throughout adult life by regulating metabolism and mitophagy

Mitch Murphy

BCH 8106

2026/09/24

https://alphataubio.com/pdf/bch8106_murphy.pdf

Draft slides were generated with Claude Sonnet 5 (extra), Gemini Pro 3.1 (extended), and ChatGPT-6 Astra (medium) with the prompt:

read this article in details every word and prepare a set of 20-25 latex beamer slides summarizing all key points in grouped subjects by figures 1-7. phd level journal club presentation for my class BCH 8106 with Dr. Mireille Khacho of uottawa. see attached sample of similar previous slides for style

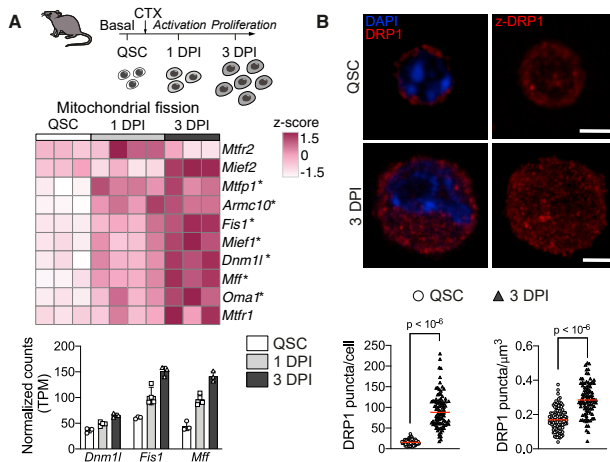
- Skeletal muscle satellite cells (SCs) maintain and regenerate muscle tissue and are normally kept in a state of quiescence (eg. reversible G0-like arrested state) throughout life.
- Upon injury, quiescent SCs (QSCs) rapidly activate and proliferate, thereby generating sufficient progeny to form new fibers and reconstruct the damaged tissue
- Proteostatic and metabolic regulation of muscle SCs is essential for maintaining their quiescence in homeostasis as well as for their capacity to activate and proliferate to regenerate damaged tissue
- Autophagy is a major proteostatic activity needed to maintain SCs in quiescence

- With aging, autophagy activity declines, leading to the accumulation of damaged cellular components (including mitochondria) and correlating with changes in metabolism
- Mitochondria have a major role in producing adenosine triphosphate (ATP) through oxidative phosphorylation (OXPHOS) via the mitochondrial electron transport chain (ETC).
- Mitochondria actively remodel their morphology via mitochondrial dynamics (fusion and fission)
- Efficiently mitochondrial fission segregates dysfunctional components of mitochondrial network and facilitates degradation via autophagy



- **KO:** *Drp1^{fl/fl}; Pax7^{CreERT2}; R26^{YFP} = *Drp1^{ΔPax7ER}*; *Drp1* mRNA ≈ 35% of WT in YFP⁺ SCs (S1D); *male**
- **Aging:** young 2.5–3 vs old 24–28 months, *female only*
- **Human:** vastus lateralis biopsies; myoblasts 25 ± 4 vs 75 ± 4 yr
- **Readouts:** deconvolved confocal + Imaris 3D surfaces; bulk RNA-seq (limma-voom, GSEA); Seahorse XF96, Oroboros O2k; mtDNA qPCR; LC3/p62/Parkin ± bafilomycin A
- **Stats:** two-tailed unpaired *t*-tests; *n* = 3–11 mice

Loss-of-function (genetic, aging) → mechanism (OXPHOS, mitophagy) → rescue by transplantation



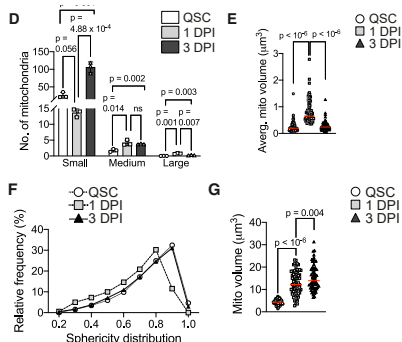
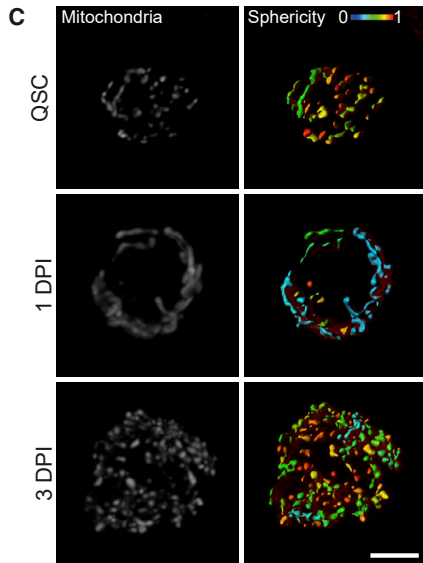
- A:** Fission-related transcripts rise during activation and proliferation, including *Dnm1l*, *Fis1* and *Mff*.
- B:** DRP1 $\uparrow$ at 3 DPI, even after normalization to cell volume.

Fission machinery rises as SCs enter proliferative phase

Satellite cell mitochondrial fission is required for muscle regeneration

Figure 1 | Remodeling changes with cell state

7/32



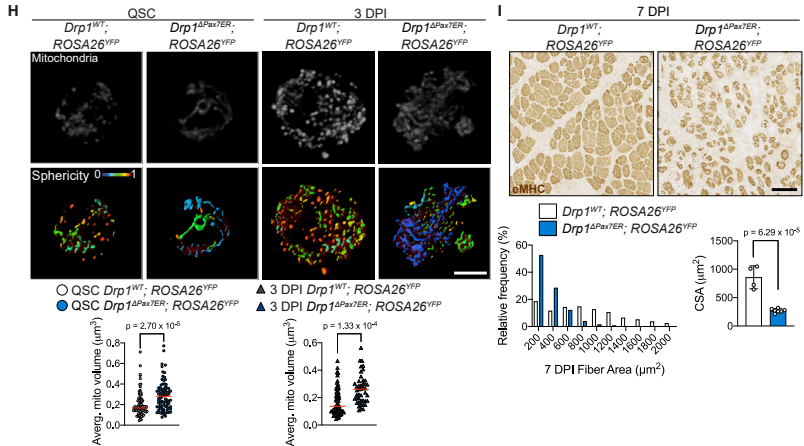
- **C–F:** At 1 DPI, mitochondria enlarge and become less spherical; at 3 DPI, small, spherical mitochondria predominate.
- **G:** At 3 DPI, individual mitochondria are smaller, but total volume per cell increases

Early fusion remodelling precedes fission proliferation

Satellite cell mitochondrial fission is required for muscle regeneration

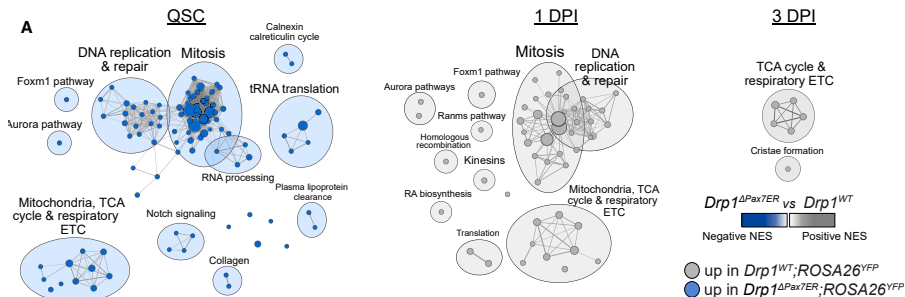
Figure 1 | DRP1 loss compromises muscle repair

8/32



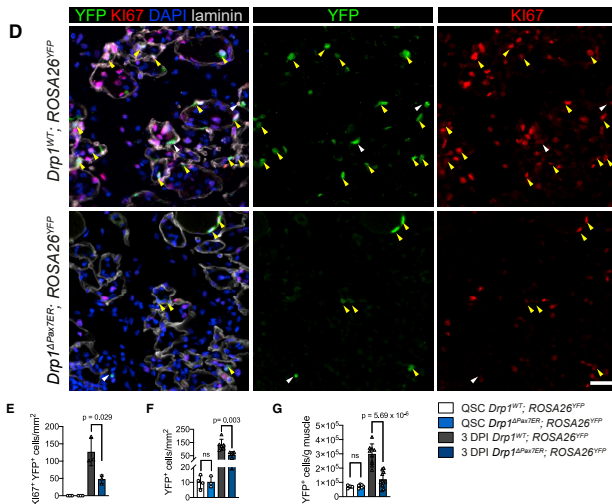
- **H:** SC-specific *Drp1* deletion produces larger, less spherical mitochondrial networks.
- **I:** *Drp1^{ΔPax7ER}* had smaller myofibers expressing embryonic myosin heavy chain (eMHC) (indicative of newly formed, regenerating myofibers) compared with WT

SC mitochondria change shape as cells progress from quiescence to proliferation and fission is essential for efficient muscle regeneration



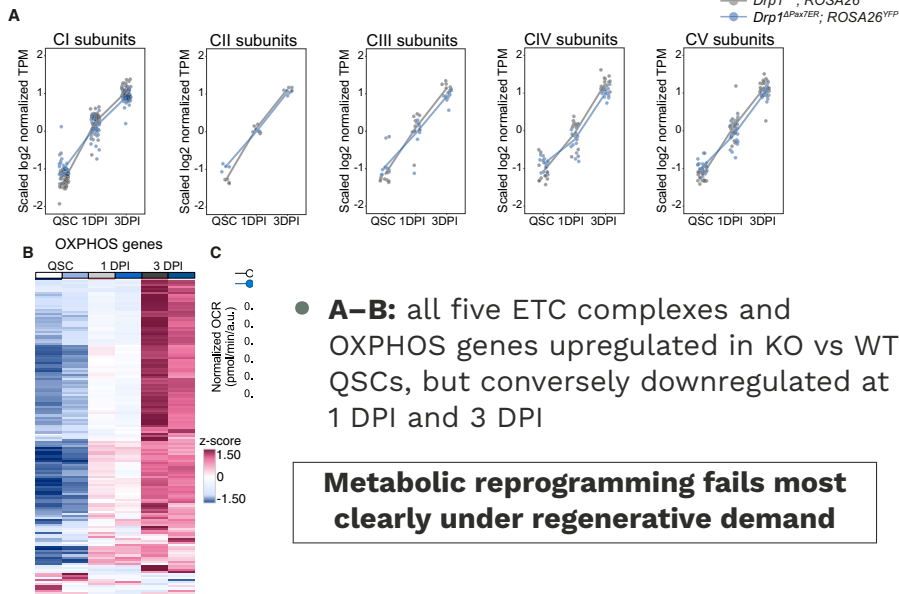
- **A–C:** Resting KO SCs express more cell-cycle and metabolic genes, yet induce these programs poorly after injury.
- A primed molecular signature does not establish proliferation or a functional G_{Alert} state.

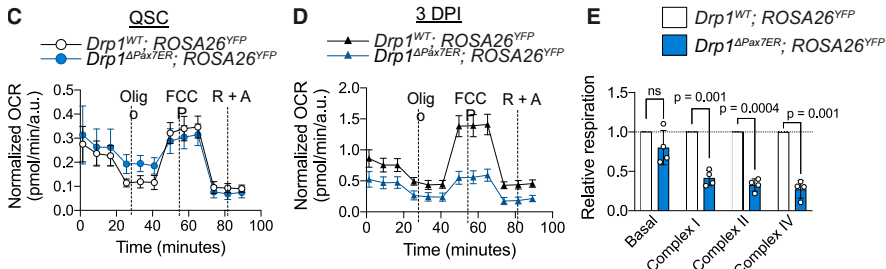
DRP1 loss uncouples molecular priming from a productive injury response



- **D–E:** YFP labels SC lineage; Ki67 marks cell cycle. At 3DPI, DRP1-deficient muscle contains fewer Ki67⁺YFP⁺ cells.
- **F–G:** Tissue counts and cell sorting similar SC numbers before injury, but fewer SCs in DRP1-deficient muscle afterward.

Activation not same as proliferation: SC can respond to injury without producing enough daughter cells for muscle repair



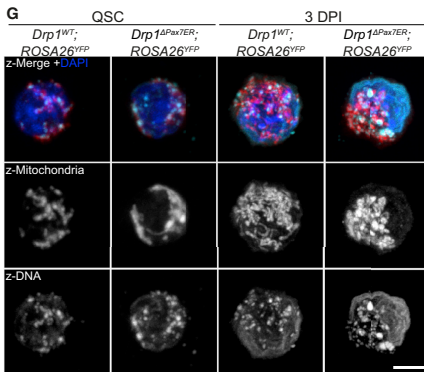
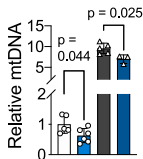


- **C–D:** Normalized oxygen-consumption rates (OCRs) at baseline and in response to oligomycin A (oligo), FCCP, or rotenone plus antimycin A (R + A)
- **C:** QSC basal and maximal respiration are similar between genotypes
- **D:** At 3 DPI, KO SCs lose maximal respiration and spare respiratory capacity.
- **E:** Cultured KO progenitors show lower complex I-, II- and IV-linked respiration.

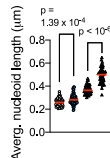
Loss of respiratory flexibility is clearer than loss of steady-state ATP

F

- QSC *Drp1*^{WT}; *ROSA26*^{YFP}
- QSC *Drp1*^{ΔPax7ER}; *ROSA26*^{YFP}
- 3 DPI *Drp1*^{WT}; *ROSA26*^{YFP}
- 3 DPI *Drp1*^{ΔPax7ER}; *ROSA26*^{YFP}



- QSC *Drp1*^{WT}; *ROSA26*^{YFP}
- QSC *Drp1*^{ΔPax7ER}; *ROSA26*^{YFP}
- ▲ 3 DPI *Drp1*^{WT}; *ROSA26*^{YFP}
- ▲ 3 DPI *Drp1*^{ΔPax7ER}; *ROSA26*^{YFP}



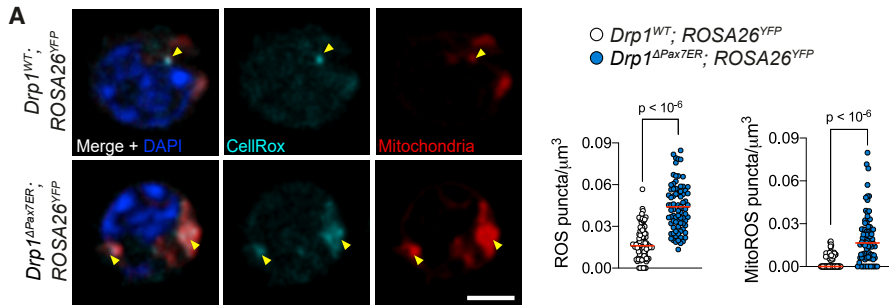
- **F:** KO SCs have fewer mtDNA copies, both at rest and after injury.
- **G:** Nucleoids are fewer and enlarged; abnormalities increase at 3 DPI.

mtDNA essential for ETC. In DRP1-deficient cells, mtDNA less abundant and abnormally organized $\Rightarrow$ reduced respiratory capacity

Fission permits mitophagy in SCs during muscle regeneration

Figure 4 | DRP1 loss increases oxidative stress

14/32



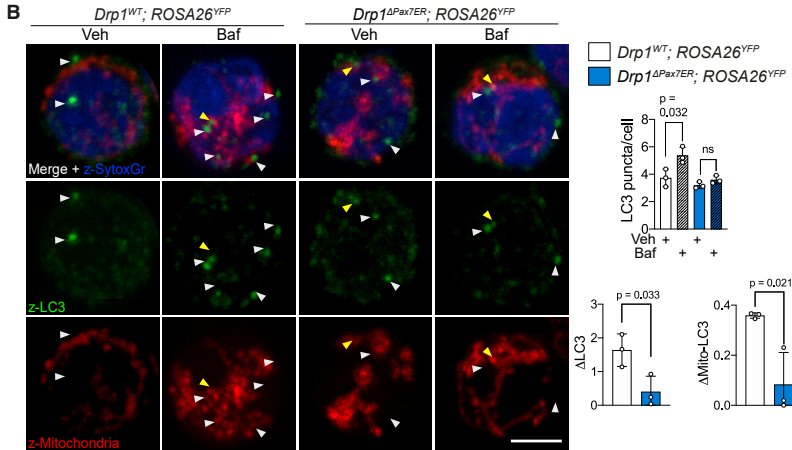
- CellROX detects oxidative stress; overlap with mitochondrial staining identifies mitochondria-associated CellROX puncta.
- DRP1-deficient SCs show increased total and mitochondria-associated CellROX puncta, normalized to cell volume.
- Loss of mitochondrial fission is accompanied by increased oxidative stress; this assay alone does not establish its source or cause.

DRP1-deficient satellite cells accumulate oxidative stress, including signals associated with mitochondria.

Fission permits mitophagy in SCs during muscle regeneration

Figure 4 | Fission loss reduces mitophagic flux

15/32

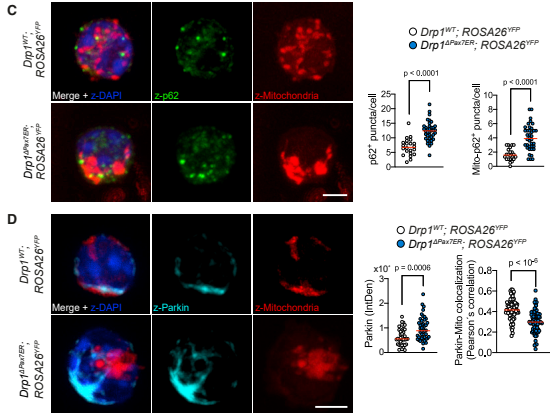


- **B:** Bafilomycin A traps material normally cleared through the lysosomal pathway
- **WT:** LC3 ↑ with blockade
- **KO:** Bafilomycin-induced increment ↓ ⇒ reduced autophagic and mitophagic flux

Blocking degradation accumulates material in LC3⁺ compartments in WT, but much less in DRP1 KO

Fission permits mitophagy in SCs during muscle regeneration

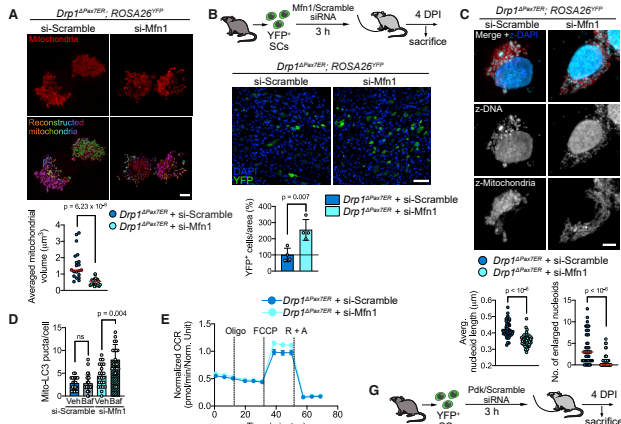
Figure 4 | p62 accumulates and Parkin localization change 16/32



- **C:** Total and mitochondrial p62 accumulate.
- **D:** Parkin ↑ but mitochondrial colocalization ↓
- p62 links ubiquitin-tagged cargo to autophagy machinery, accumulation around mitochondria with reduced mitophagic flux supports impaired clearance

- Parkin tags mitochondrial proteins with ubiquitin for mitophagy. Despite increased abundance, reduced mitochondrial colocalization suggests impaired recruitment

Without DRP1, satellite cells accumulate oxidative stress and show signs of impaired mitochondrial clearance.

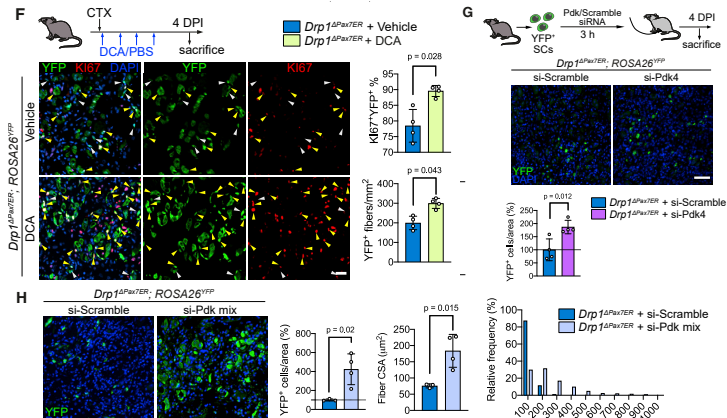


- **A–C:** *Mfn1* silencing reduces mitochondrial size and nucleoid aggregation and improves donor-cell accumulation.

- **D–E:** Mitophagic flux and maximal respiration improve.

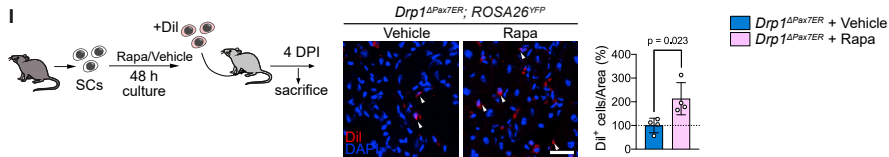
- Without DRP1, mitochondria cannot divide normally but continue to fuse
- Reducing MFN1 limits fusion and improves mitochondrial function even though DRP1 remains absent

Reducing fusion improves mitochondrial function and SC expansion when mitochondrial division impaired



- **F:** 4D dichloroacetate (DCA) ↑ cycling SCs and regenerating YFP⁺ fibers.
- **G–H:** *Pdk4* silencing improves donor-cell accumulation; pooled *Pdk1–4* silencing also improves fiber size.
- PDK inhibition favors pyruvate oxidation through PDH and TCA cycle

Independent metabolic interventions improve regeneration

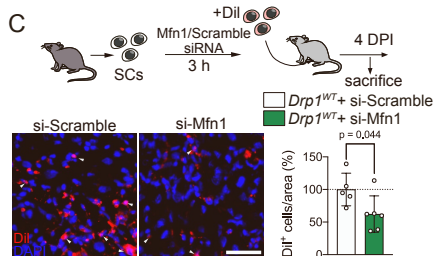


- **I:** Pretreat KO donor SCs with rapamycin for 48 h, label with Dil and transplant.
- **Result:** More Dil⁺ donor cells are detected in recipient muscle four days later.
- **Conclusion:** A proteostasis-directed intervention can improve early engraftment.
- **Caution:** Rapamycin has broad mTOR effects; this does not isolate mitophagy as the sole mediator.

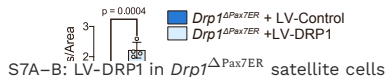
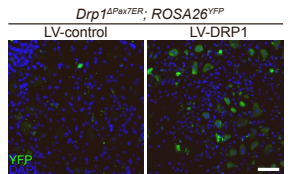
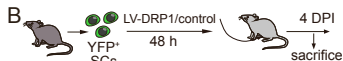
Restoring function does not require re-expressing DRP1 in every experiment

Transient adynamism restores proliferative capacity of Drp1-null SCs

Figs. S5C, S7A–B | Specificity control and genetic rescue 20/32



S5C: si-*Mfn1* in $Drp1^{WT}$ satellite cells



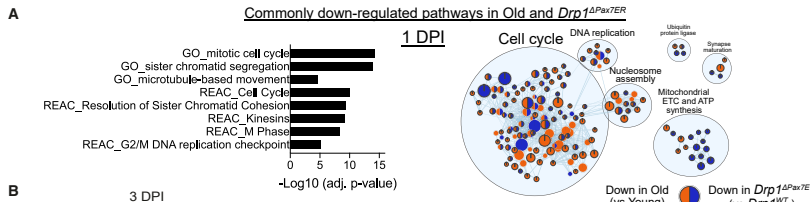
- **S5C — the specificity control.** Silencing *Mfn1* in *wild-type* SCs reduces engraftment ($\approx 60\%$ of scramble, $p = 0.044$). So less fusion is not good in itself: what the rescue restores is the fusion/fission *balance*.
- **S7A–B — closing the genetic loop.** Lentiviral DRP1 (hPGK-*DNM1L*) restores DRP1 protein and rescues $Drp1^{\Delta Pax7ER}$ SC expansion ($\approx 2.2\times$, $p = 0.0004$). The phenotype is caused by loss of DRP1 itself, not by unrelated effect of floxed allele or tamoxifen.

Re-expressing DRP1 rescues the knockout; blocking fusion helps only when fission is already lost

Impaired fission causes regenerative failure in old SCs

Figure 6 | Do old SCs resemble cells lacking DRP1?

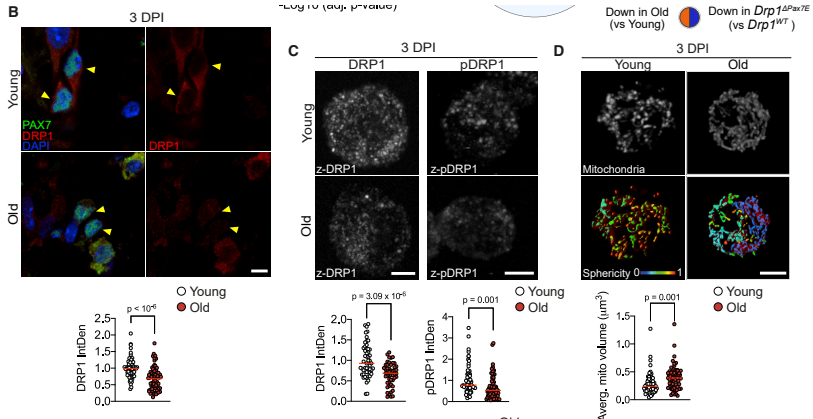
21/32



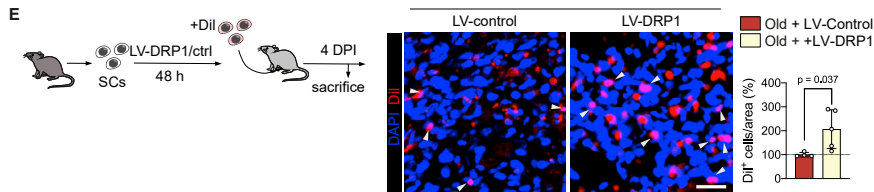
- **Question:** Earlier figures showed that deleting DRP1 impairs SC proliferation and mitochondrial function. Do similar changes occur during normal aging?
- **Comparison:** The authors compare gene expression in old versus young SCs, and separately in DRP1-null versus control SCs, at 1 day after injury.
- **Result:** Both comparisons show reduced expression of groups of genes involved in cell division and mitochondrial energy production. These functional gene groups are called *pathways*.
- **Interpretation:** Old SCs share some molecular defects with DRP1-null SCs. This similarity motivates testing whether DRP1 itself is reduced with aging.

Impaired fission causes regenerative failure in old SCs

Figure 6 | Old SCs have less DRP1 and larger mitochondria 22/32



- **Question:** Is the machinery that divides mitochondria reduced in old SCs during the proliferative phase, 3 days after injury?
- **B:** In muscle sections, PAX7 identifies satellite cells. These cells show less DRP1 staining in old mice than in young mice.
- **C:** Isolated old SCs also show less total DRP1 and less phosphorylated DRP1 (pDRP1) than young SCs, consistent with impairment of mitochondrial fission.



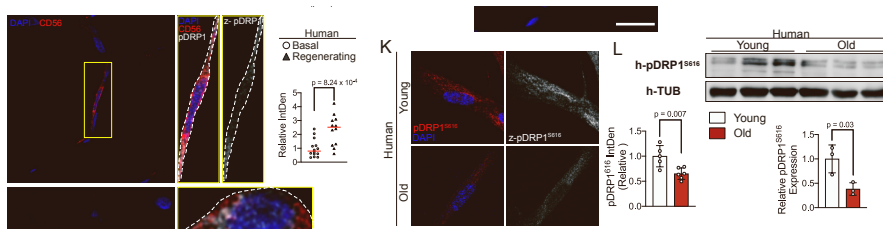
- **E:** Old donor SCs receive lentiviral DRP1 or a control vector before transplantation.
- **Result:** DRP1 expression increases Dil⁺ donor-cell accumulation at the four-day endpoint.
- DRP1 availability is functionally limiting in these conditions.

Aging association strengthened by direct rescue experiment

Impaired fission causes regenerative failure in old SCs

Figures S6I–L | Does any of this hold in humans?

24/32

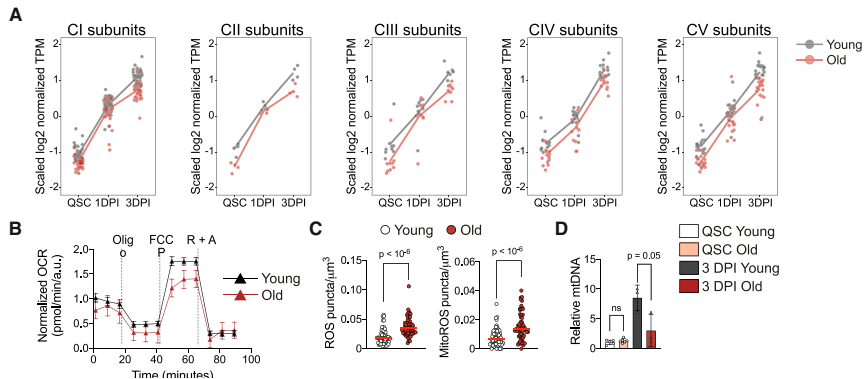


S6I: human muscle biopsies, CD56⁺ SCs

S6K–L: young vs aged human myoblasts

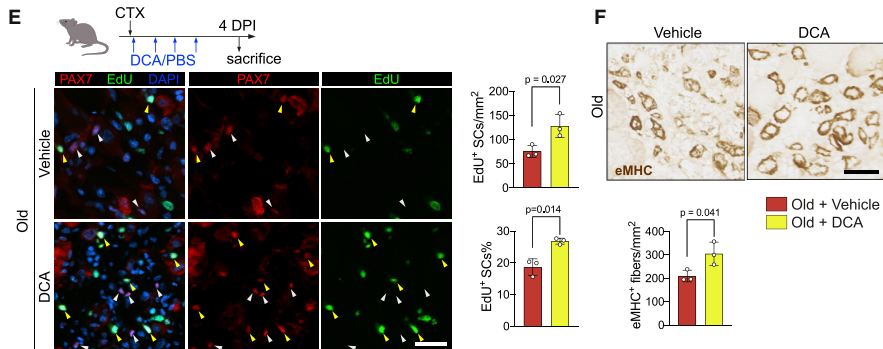
- **S6I:** In biopsies, CD56⁺ SCs with high active pDRP1^{S616} are found in *regenerating* but not intact regions ($p = 8.2 \times 10^{-4}$) — fission activity is induced by damage in human muscle too.
- **S6K–L:** Myoblasts from aged donors (75±4 yr) vs young (25±4 yr) have less pDRP1^{S616} by imaging ($\approx 0.65\times$, $p=0.007$) and by western blot ($\approx 0.4\times$, $p=0.03$).
- **S6J:** Aged human myoblasts proliferate less: BrdU⁺ falls from $\approx 34\%$ to $\approx 10\%$ ($p=0.001$), mirroring the mouse phenotype.
- **Scope:** $n = 3$ young and 4 aged donors, correlative only — no human gain-of-function experiment. Donor sex and comorbidities are not reported.

The fission–aging link present in human SCs, but only as correlation



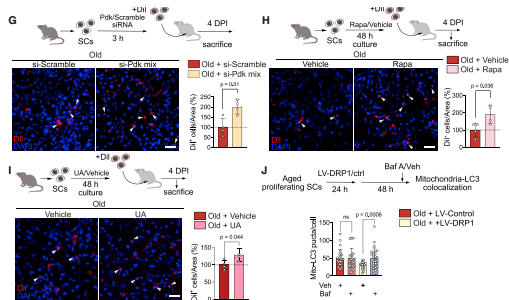
- **A–B:** ETC gene induction and respiratory performance are lower in old SCs.
- **C:** Oxidative-stress signals rise despite reduced OXPHOS capacity.
- **D:** Injury-induced mtDNA expansion is about 10-fold in young SCs versus 2–3-fold in old SCs.

Old SCs resemble stressed fission-impaired state

Figure 7 | OXPHOS support improves aged muscle repair 26/32

- **E–F:** DCA increases PAX7⁺EdU⁺ SCs and the number of eMHC⁺ fibers.
- **G:** Pooled *Pdk* silencing in old donor cells improves early accumulation after transplantation.
- Aged phenotype retains a reversible metabolic component.

Pharmacological and donor-cell genetic interventions converge



- **H–I:** Rapamycin or urolithin A pretreatment increases Dil⁺ donor-cell accumulation.
- **J:** DRP1 re-expression increases the bafilomycin-dependent accumulation of mitochondrial LC3.
- Restored DRP1 and mitophagic throughput are linked.
- Does genetic blockade of mitophagy abolish the DRP1 rescue?

Mitophagy remains a testable mediator of DRP1-dependent regeneration

Satellite cell mitochondrial fission is required for muscle regeneration

Mitochondrial fission permits satellite cell expansion after tissue injury

Fission allows SCs to induce OXPHOS metabolism during muscle regeneration

Fission permits mitophagy in SCs during muscle regeneration

Transient adynamism restores proliferative capacity of Drp1-null SCs

Impaired fission causes regenerative failure in old SCs

Restoring OXPHOS and mitophagy rescues regenerative failure of fission-impaired aged SCs

BONUS

- Only DRP1 experimented ? What about FIS1, MFF, ... (further studies ?)
- “*Representative IF images*” all images should be disclosed in supplemental to demonstrate no selection bias
- “*only old female mice were used, due to the limited availability of old male mice*”
 - **Sex-Dependent Gene Expression:** The expression and activity of several of these mitochondrial fission genes—most notably Dnm1l (which encodes DRP1), Fis1, and Mff—are highly sex-dependent.
 - **Hormonal Regulation:** Mitochondrial dynamics (balance between fusion and fission) are heavily regulated by sex hormones. Estrogen is a major regulator of mitochondrial biogenesis and function in skeletal muscle; it directly modulates DRP1 activity, influences how docking proteins like Fis1 and Mff recruit DRP1 to the mitochondrial membrane, and alters baseline oxidative phosphorylation.
- **what other limitations did you identify ?**

- “Common Errors in Statistics”
underlying fundamentals causing crisis of reproducibility in bioinformatics
- youtube electron transport chain and atp synthase
- youtube sandia lab

